

Table 2: The performance of Scatterbrain, Reformer, Performer and Full-Attention on Long-Range-Arena benchmarks and 2 popular language modeling tasks. We fix the same number of parameters (1/8 of the full) used for approximating the attention matrix for each method.

Attention	Copy (ppl)	WikiText-103 (ppl)
Full Attention	1	25.258
Reformer	6.8	27.68
Performer	49	66
Scatterbrain	2.58	26.72

Attention	ListOps	Text	Retrieval	Image	Pathfinder	Avg
Full Attention	38.2	63.29	80.85	41.78	73.98	59.62
Reformer	36.85	58.12	78.36	28.3	67.95	53.9
Performer	35.75	62.36	78.83	39.71	68.6	57.05
Scatterbrain	38.6	64.55	80.22	43.65	69.91	59.38

Drop-in replacement for full attention: We show that accurate approximation directly leads to efficient Inference. We replace BigGAN’s dense attention with a Scatterbrain layer without other modifications. In 5 (right two), we show Inception and FID scores for Scatterbrain and other baselines under different memory budgets. Similarly, we use T2T-ViT [69], which is a token-to-token vision Transformer pre-trained on ImageNet [25]. In Table 1, we show the average approximation error of Scatterbrain for each layer and the end-to-end testing accuracy after replacing full attention with Scatterbrain and other baselines. Notably, Scatterbrain achieves 80.7% Top-1 accuracy, which is only 1% drop from the original 81.7% by full attention reducing up to 98% of the memory usage.

5.2 End-to-end Training Performance

Scatterbrain’s accurate approximation of attention matrices allows it to outperform other efficient Transformer methods on benchmark tasks. Across a range of diverse tasks, both commonly used autoregressive tasks (sequence modeling) and benchmark long-range classification tasks (Long-Range Arena), Scatterbrain outperforms Performer (low-rank baseline) and Reformer (sparse baseline) by up to 4 points.

5.2.1 Auto-regressive Tasks

On the standard language modeling task of Wikitext-103, Scatterbrain obtains 1 point better perplexity than Reformer (sparse baseline), coming within 1.5 points of full attention.

Settings: We compare the performance of Scatterbrain against Reformer and Performer on one popular synthetic task, Copy, and one large language modeling task: WikiText103 [45]. Reformer is a representative sparse-approximation-based variant and Performer is a low-rank-approximation-based variant. The base model is vanilla Transformer [63]. We observed that generally allocating more memory budget to sparse tends to perform better, so Scatterbrain sets the ratio to 3:1 (sparse: low-rank component) for simplicity. The statistics of each dataset and model hyper-parameters are in Appendix E. We report the best results of each method in perplexity.

Results: Table 2 shows the testing perplexity for Scatterbrain and other baselines under the same parameter budget (each approximation is only allowed to compute $\frac{1}{8}$ of the full computation). Scatterbrain achieves comparable perplexity compared to the full attention Transformer model on Copy, and WikiText-103. Notably, Scatterbrain achieves 4 points lower perplexity on Copy and 1 point lower on WikiText-103 compared to Reformer, while Performer does not train stably on auto-regressive tasks (loss does not go down).

Analysis: We also analyze the results by visualizing the error of Reformer (sparse), Performer (low-rank), and Scatterbrain (sparse + low-rank) given the same number of parameters when approximating the full attention matrices for each attention layer during training (Appendix E). The conclusion is for language modeling tasks, sparse+low-rank has the smallest approximation error in most of the cases, and sparse has the largest error, which matches with the end-to-end results. It also confirms the observation in the popular benchmark paper [57] that kernel or low-rank based approximations are less effective for hierarchical structured data.

5.2.2 Classification Tasks

On a suite of long-range benchmark tasks (Long Range Area), Scatterbrain outperforms Reformer (sparse baseline) and Performer (low-rank baseline) by up to 5 points on average.

Settings: We compare the performance of Scatterbrain against Reformer and Performer on ListOps, two classifications: byte-level IMDb reviews text classification, image classification on sequences of pixels, a text retrieval, and pathfinder tasks. The datasets are obtained from the Long Range Arena (LRA) Benchmark [57], which is a recent popular benchmark designed for testing efficient Transformers. Similar to the auto-regressive tasks above, we use Reformer and Performer as baselines. The base model is also a vanilla Transformer. We follow the evaluation protocol from [57]. We report the best accuracy of each method.

Results: Table 2 shows the individual and average accuracy of each task for Scatterbrain and other baselines under the same parameters budget. Specially, each approximation is only allowed to use 12.5% of the full computation. We can see Scatterbrain is very close to full attention even with a large reduction in computation and memory. Further more, it outperforms all the other baselines consistently on every task and achieves more than 5 point average accuracy improvement than sparse-based approximation Reformer and more than 2 point average accuracy improvement than low-rank-based variant Performer.

Analysis: Similarly, in order to analyze the performance of Reformer, Performer and Scatterbrain, we visualize their approximation error given the same number of parameters when approximating the full attention matrices for each attention layer during training (Appendix E). We again find that Scatterbrain has the smallest approximation error, while Performer is the worst on ListOps and Reformer has the largest error on classification tasks, which matches with the end-to-end results and confirms our observations earlier (sparse and low-rank approximation excel in different regimes).

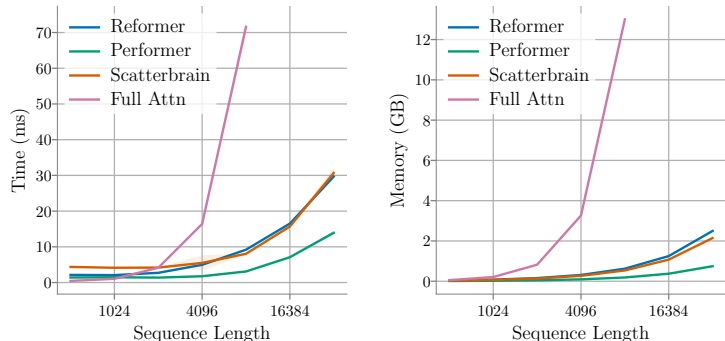


Figure 6: Speed and memory required by different efficient attention methods. Scatterbrain is competitive with SMYRF (sparse baseline) and Performer (low-rank baseline), while up to $3\times$ faster and $12\times$ more memory efficient than full attention for sequence length 4096.

5.3 Scatterbrain’s Efficiency, Scaling with Input Sequence Length

We include ablation studies on the scalability of Scatterbrain in Fig. 6, showing that it is as computation and memory-efficient as simpler baselines such as SMYRF and Performer, while up to $3\times$ faster and $12\times$ more memory efficient than full attention for sequence length 4096. This demonstrates that our combination of sparse and low-rank inherits their efficiency.

We report run times and memory consumption of the sequence lengths ranging from 512 to 32768. We use a batch size of 16 for all runs and conduct experiments a V100 GPU. Since the efficiency would be largely conditioned on hardware and implementation details, we perform best-effort fair comparisons. We adapt the Pytorch implementation from `pytorch-fast-transformers` library for our baselines and implement Scatterbrain similarly without any customized cuda kernels.

6 Discussion

Limitations. As Scatterbrain has sparse attention as a component, it is not yet as hardware friendly (on GPUs and TPUs) as the low-rank component, which uses the very optimized dense matrix multiplication. This is the same limitation suffered by other sparse attention methods, but we are excited that more efficient sparse GPU kernels are being developed [29, 31].

Potential negative societal impacts. Our work seeks to understand the role of matrix approximation (and potentially energy savings) in the attention layer, which may improve a wide range of applications, each with their own potential benefits and harms. For example, making it language modeling more compute and memory efficient might facilitate spreading misinformation, and better image and video processing may make automatic surveillance easier. To mitigate these risks, one needs to address application-specific issues such as privacy and fairness, going beyond the error metrics we considered. Specially, for language models (LMs), while our work partially addresses the issue of environmental cost of LMs raised in [6], it does not address other issues such as unfathomable training data [6].

Discussion and future work. In this work, we make an observation on the sparse + low-rank structure of the attentions in Transformer models and theoretically characterize the regimes where sparse, low-rank and sparse + low-rank excel, based on the softmax temperature of the attention matrices. Motivated by this observation, we present Scatterbrain, a novel way to unify the strengths of both sparse and low-rank methods for accurate and efficient attention approximation with provable guarantees. We empirically verify the effectiveness of Scatterbrain on pretrained BigGAN, vision transformers, as well as end-to-end training of vanilla transformer. We anticipate that the study of this core approximation problem can prove useful in other contexts, such as generalized attention layers with other non-linearity beside softmax, and wide output layer in language modeling or extreme-classification.

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References

- [1] Keivan Alizadeh, Ali Farhadi, and Mohammad Rastegari. Butterfly transform: An efficient FFT based neural architecture design. In *The Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [2] Alexandr Andoni, Piotr Indyk, Thijs Laarhoven, Ilya Razenshteyn, and Ludwig Schmidt. Practical and optimal lsh for angular distance. In C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, editors, *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1225–1233. 2015.
- [3] Alexandr Andoni, Piotr Indyk, Thijs Laarhoven, Ilya Razenshteyn, and Ludwig Schmidt. Practical and optimal LSH for angular distance. In *Proceedings of the 28th International Conference on Neural Information Processing Systems-Volume 1*, pages 1225–1233, 2015.
- [4] R Artusi, P Verderio, and E Marubini. Bravais-pearson and spearman correlation coefficients: meaning, test of hypothesis and confidence interval. *The International journal of biological markers*, 17(2):148–151, 2002.

- [5] Iz Beltagy, Matthew E Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.
- [6] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Margaret Mitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, New York, NY, USA, 2021. Association for Computing Machinery.
- [7] Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020.
- [8] Emmanuel J Candès and Benjamin Recht. Exact matrix completion via convex optimization. *Foundations of Computational mathematics*, 9(6):717–772, 2009.
- [9] Emmanuel J Candès, Xiaodong Li, Yi Ma, and John Wright. Robust principal component analysis? *Journal of the ACM (JACM)*, 58(3):1–37, 2011.
- [10] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision*, pages 213–229. Springer, 2020.
- [11] Beidi Chen and Anshumali Shrivastava. Densified winner take all (wta) hashing for sparse datasets. In *Uncertainty in artificial intelligence*, 2018.
- [12] Beidi Chen, Anshumali Shrivastava, and Rebecca C Steorts. Unique entity estimation with application to the syrian conflict. *The Annals of Applied Statistics*, 12(2):1039–1067, 2018.
- [13] Beidi Chen, Yingchen Xu, and Anshumali Shrivastava. Fast and accurate stochastic gradient estimation. 2019.
- [14] Beidi Chen, Tharun Medini, James Farwell, Charlie Tai, Anshumali Shrivastava, et al. SLIDE: In defense of smart algorithms over hardware acceleration for large-scale deep learning systems. *Proceedings of Machine Learning and Systems*, 2:291–306, 2020.
- [15] Beidi Chen, Zichang Liu, Binghui Peng, Zhaozhuo Xu, Jonathan Lingjie Li, Tri Dao, Zhao Song, Anshumali Shrivastava, and Christopher Ré. Mongoose: A learnable lsh framework for efficient neural network training. In *The International Conference on Learning Representations (ICLR)*, 2021.
- [16] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating long sequences with sparse transformers. *arXiv preprint arXiv:1904.10509*, 2019.
- [17] Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, et al. Rethinking attention with performers. *arXiv preprint arXiv:2009.14794*, 2020.
- [18] Shabnam Daghighi, Tharun Medini, Nicholas Meisburger, Beidi Chen, Mengnan Zhao, and Anshumali Shrivastava. A tale of two efficient and informative negative sampling distributions. In *International Conference on Machine Learning*, pages 2319–2329. PMLR, 2021.
- [19] Zihang Dai, Zhilin Yang, Yiming Yang, Jaime Carbonell, Quoc V Le, and Ruslan Salakhutdinov. Transformer-xl: Attentive language models beyond a fixed-length context. *arXiv preprint arXiv:1901.02860*, 2019.
- [20] Tri Dao, Albert Gu, Matthew Eichhorn, Atri Rudra, and Christopher Ré. Learning fast algorithms for linear transforms using butterfly factorizations. In *The International Conference on Machine Learning (ICML)*, 2019.